

Neurons to networks

1 A neuron is logistic regression

Weigh the inputs, add them up, squash the total. Part I already built one.

2 One neuron draws a straight line

It can't separate "unusual in any direction". A few hidden neurons can bend the line.



3 Hinges make any shape

ReLU is a hinge. Add enough hinges, and a network can draw almost any curve. Without them, layers collapse into one line.

4 Backpropagation shares the blame

Same gradient descent as Chapter 3. The error flows backwards, layer by layer, telling each weight which way to move.

5 Deep = many layers

Each layer builds features from the one before. On raw pixels and words, that's what makes deep learning work.

6 Bigger isn't automatically better

On Kestrel's neat features, logistic regression scored 0.175; an oversized network 0.511, and it was overconfident.